US012403493B2

(12) **United States Patent**
Hang et al.

(10) **Patent No.: US 12,403,493 B2**
(45) **Date of Patent: Sep. 2, 2025**

(54) **ALL-PLASTIC SPRAY GUN**

(71) Applicant: **NINGBO JINYU TECHNOLOGY INDUSTRY CO., LTD.**, Ningbo (CN)

(72) Inventors: **Cheng Hang**, Ningbo (CN); **Gongshun Duan**, Ningbo (CN)

(73) Assignee: **NINGBO JINYU TECHNOLOGY INDUSTRY CO., LTD.**, Ningbo (CN)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 208 days.

(21) Appl. No.: **18/373,978**

(22) Filed: **Sep. 28, 2023**

(65) **Prior Publication Data**

US 2025/0058340 A1 Feb. 20, 2025

Related U.S. Application Data

(63) Continuation of application No. PCT/CN2023/115077, filed on Aug. 17, 2023.

(51) **Int. Cl.**
B05B 11/10 (2023.01)

(52) **U.S. Cl.**
CPC ***B05B 11/1077*** (2023.01); ***B05B 11/1009*** (2023.01)

(58) **Field of Classification Search**
CPC B05B 11/1077; B05B 11/1009; B05B 11/1011; B05B 11/00; B05B 11/0005; B05B 11/025
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,169,032 A * 12/1992 Steijns B05B 11/0029 222/153.07
5,228,600 A * 7/1993 Steijns B05B 1/12 222/153.05
6,641,003 B1 * 11/2003 Foster B05B 11/007 222/383.1
2006/0086762 A1 * 4/2006 Foster B05B 11/1087 222/383.1
2010/0200675 A1 * 8/2010 Tu B05B 11/1011 239/428.5
2025/0058340 A1 * 2/2025 Hang B05B 11/1011

FOREIGN PATENT DOCUMENTS

CN 215235147 U * 12/2021

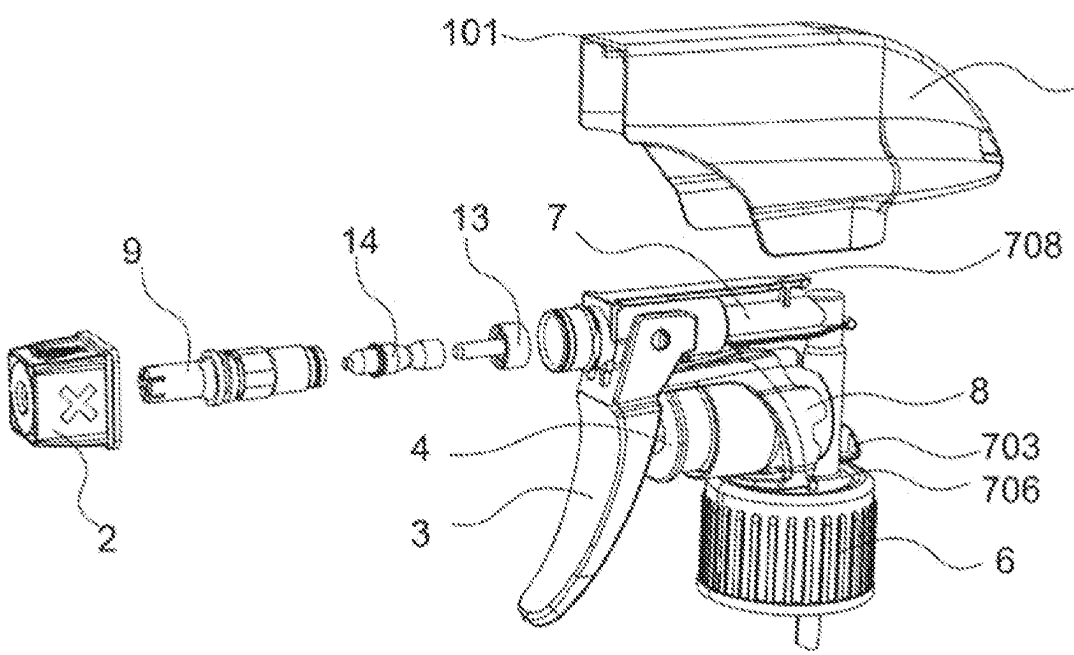
* cited by examiner

Primary Examiner — Bob Zadeh
(74) *Attorney, Agent, or Firm* — Jose Cherson Weissbrot

(57) **ABSTRACT**

Disclosed is an all-plastic spray gun, including a housing. The housing is connected to a gun body, a front end of the gun body is connected to a header cap, the header cap is connected to a water division head body, the water division head body is inserted into the gun body, the gun body is hinged to a trigger, the trigger is connected to a piston, the piston is slidably connected to a negative pressure column of the gun body, an upper end of the negative pressure column is provided with a plastic spring, the plastic spring is sleeved outside the gun body, a bottom end of the gun body is connected to a blank cap, the blank cap is connected to a large ring, a perforative suction tube is inserted into the blank cap.

10 Claims, 14 Drawing Sheets



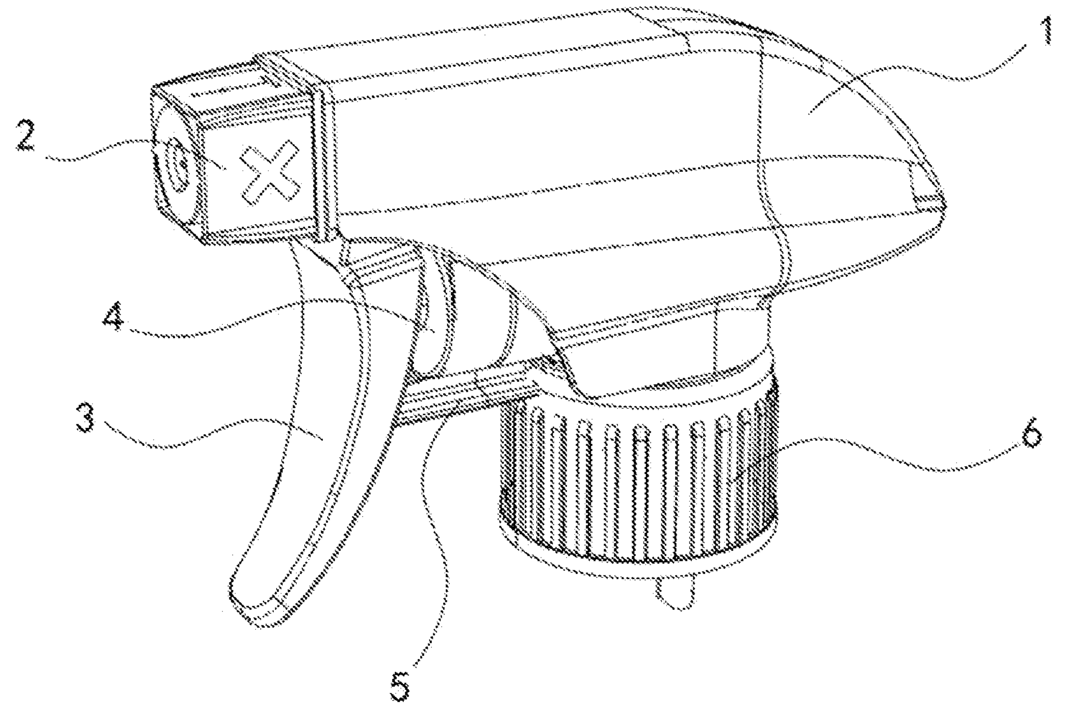


FIG. 1

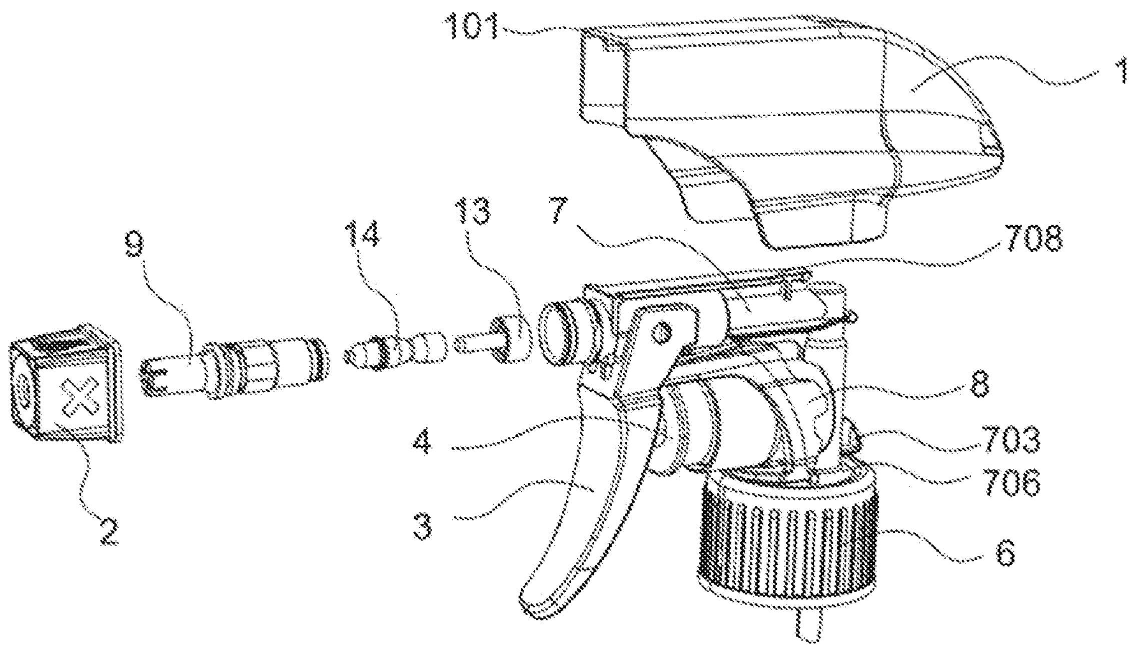


FIG. 2

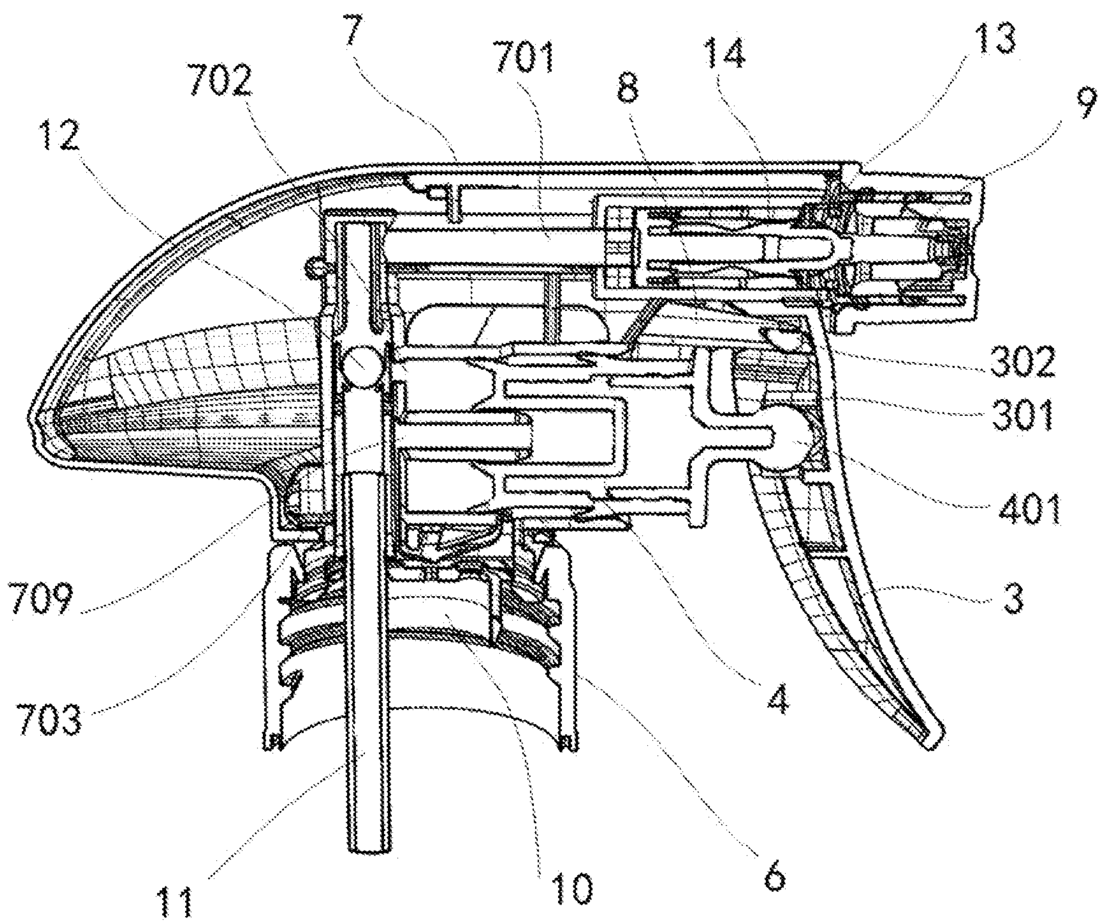


FIG. 3

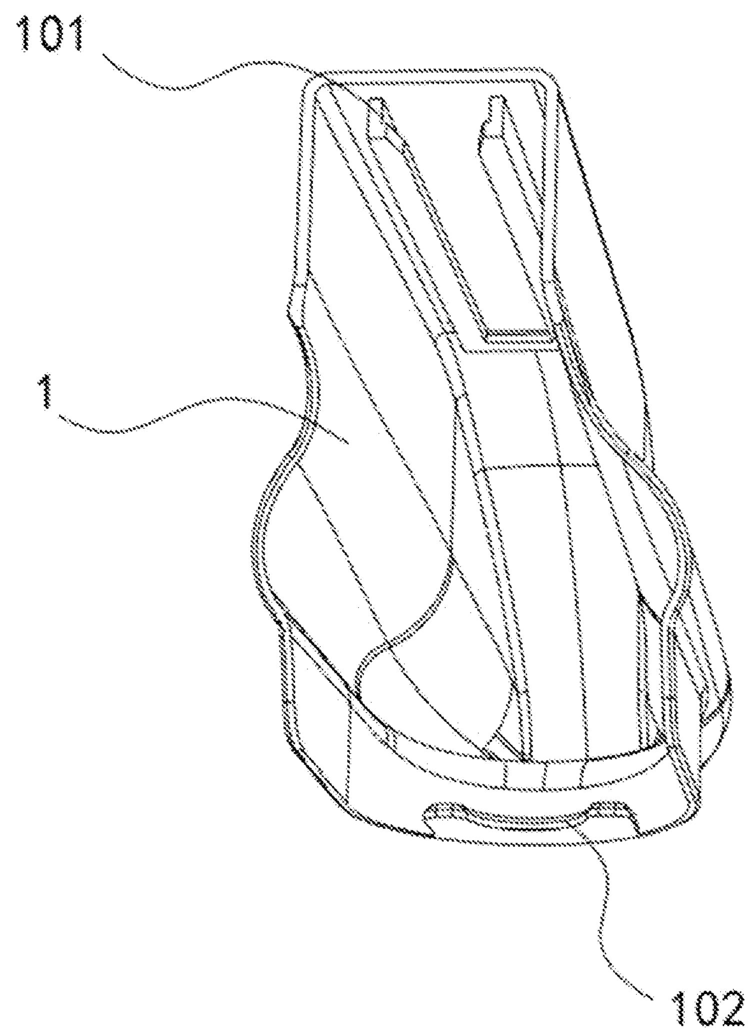


FIG. 4

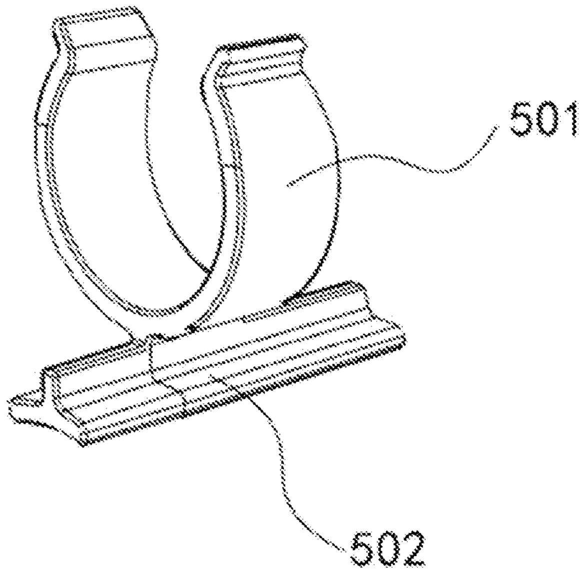


FIG. 5

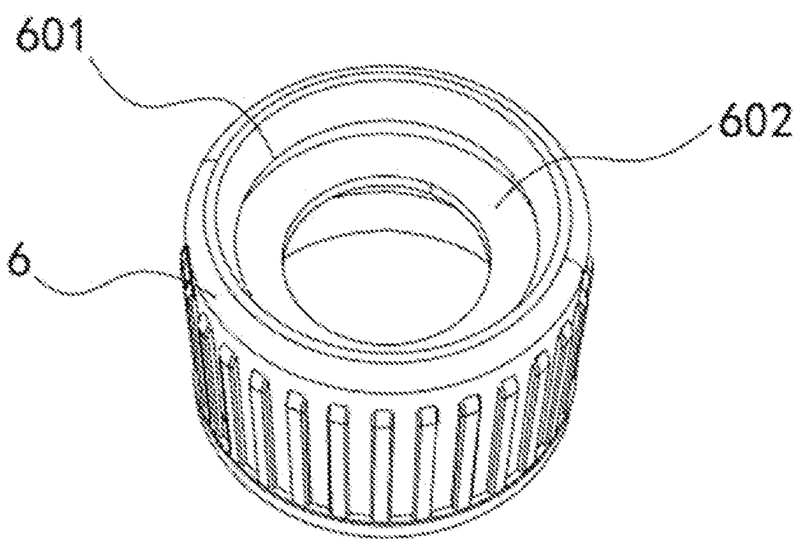


FIG. 6

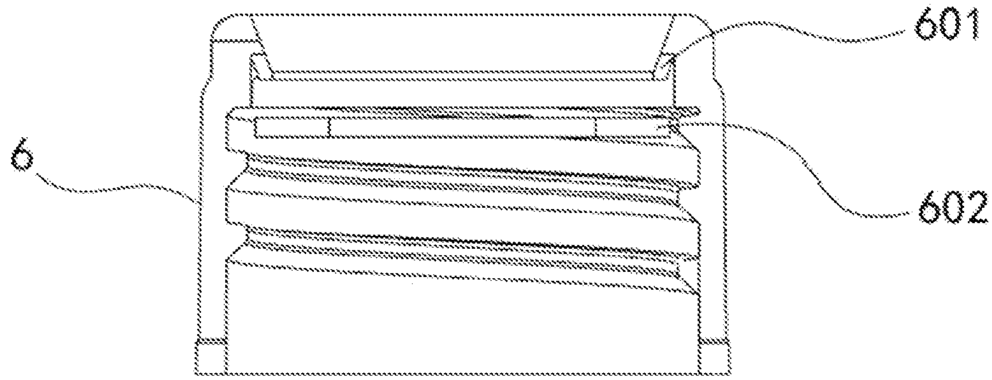


FIG. 7

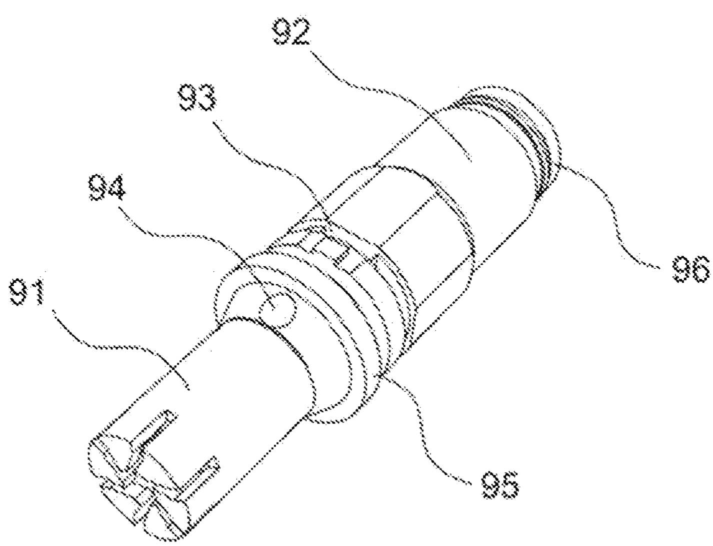


FIG. 8

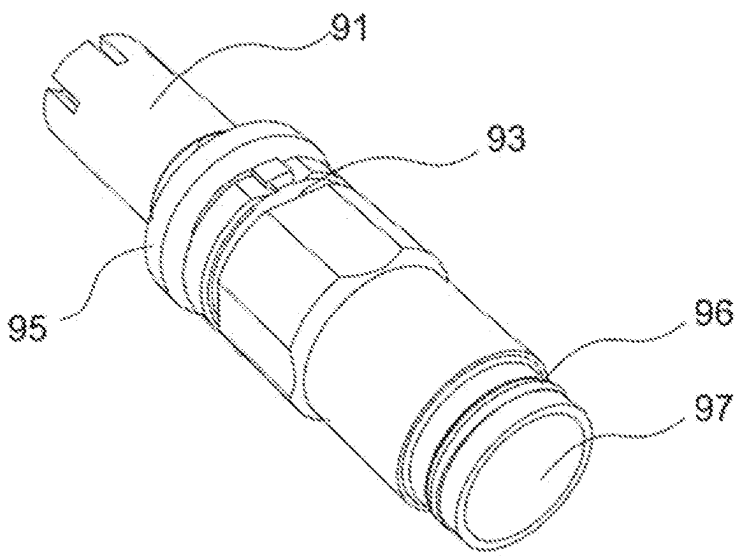


FIG. 9

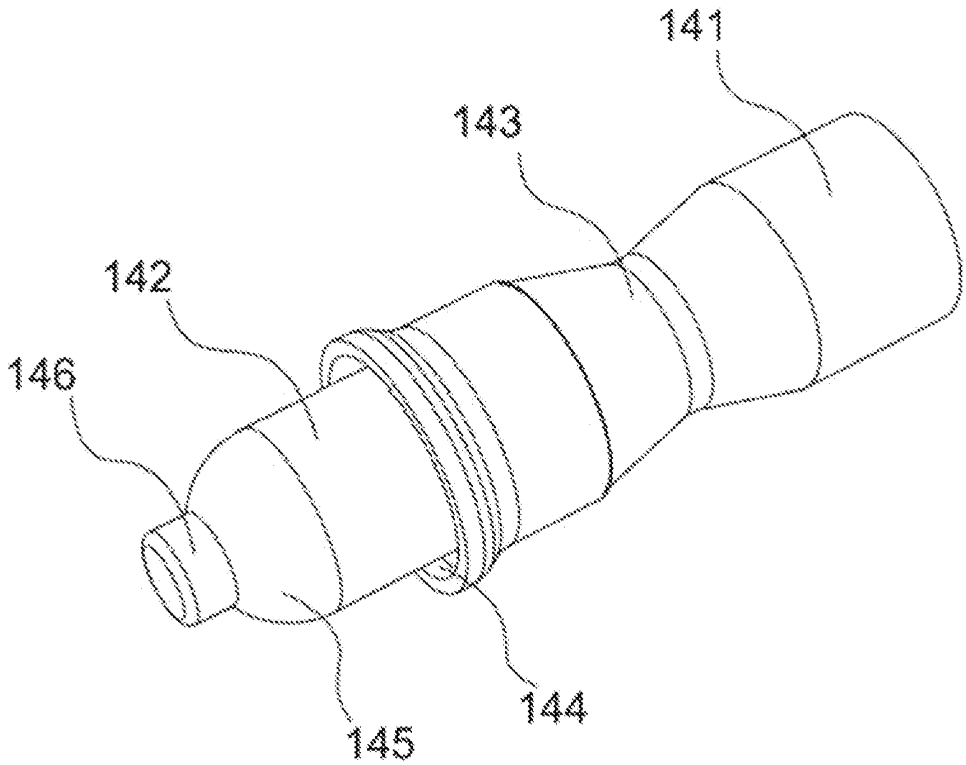


FIG. 10

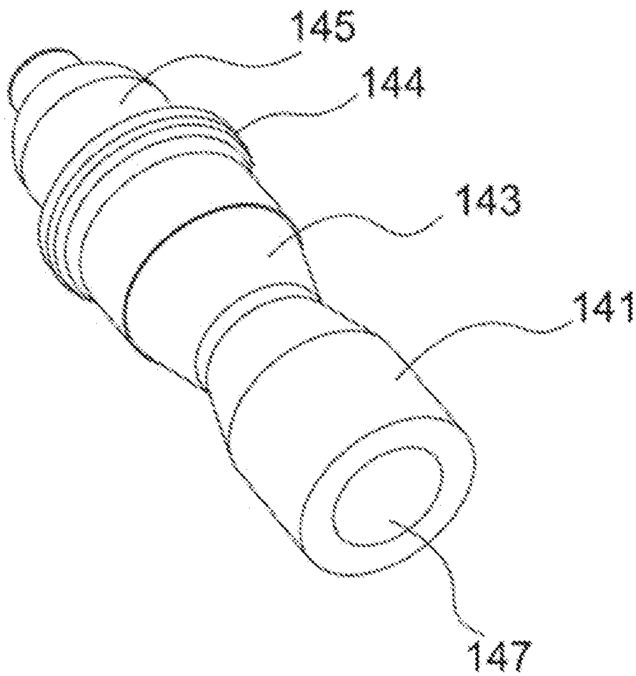


FIG. 11

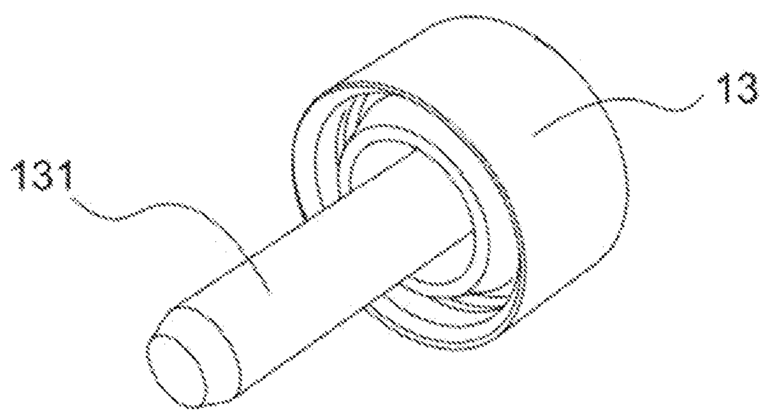


FIG. 12

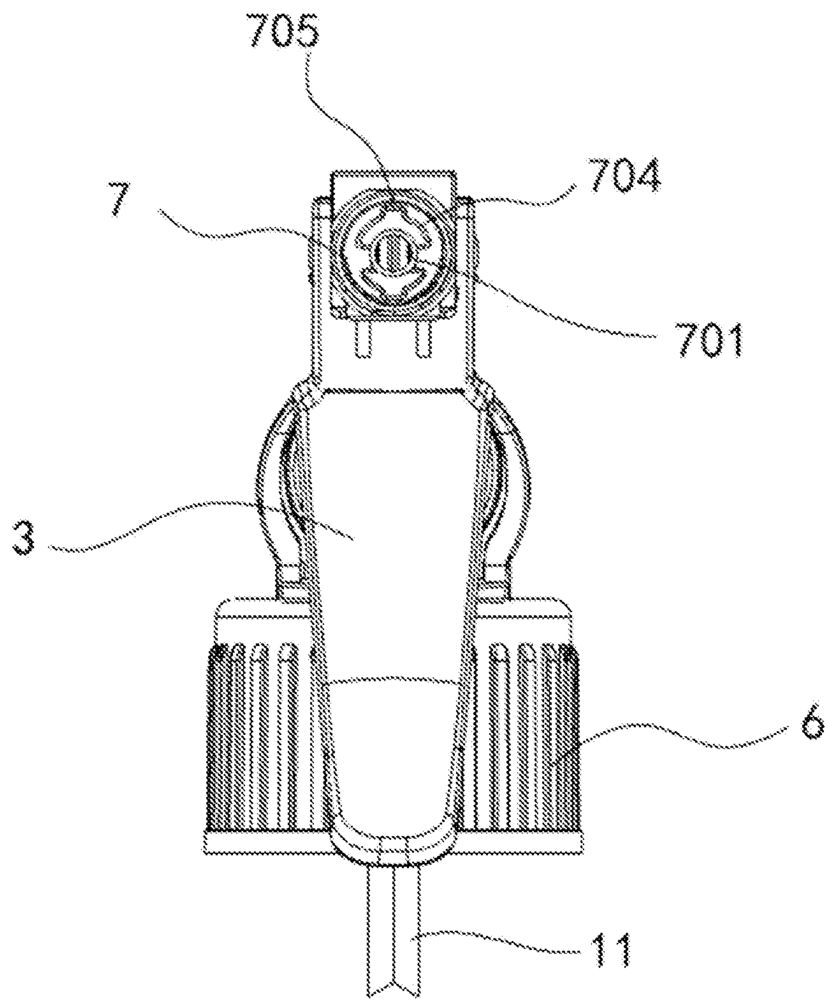


FIG. 13

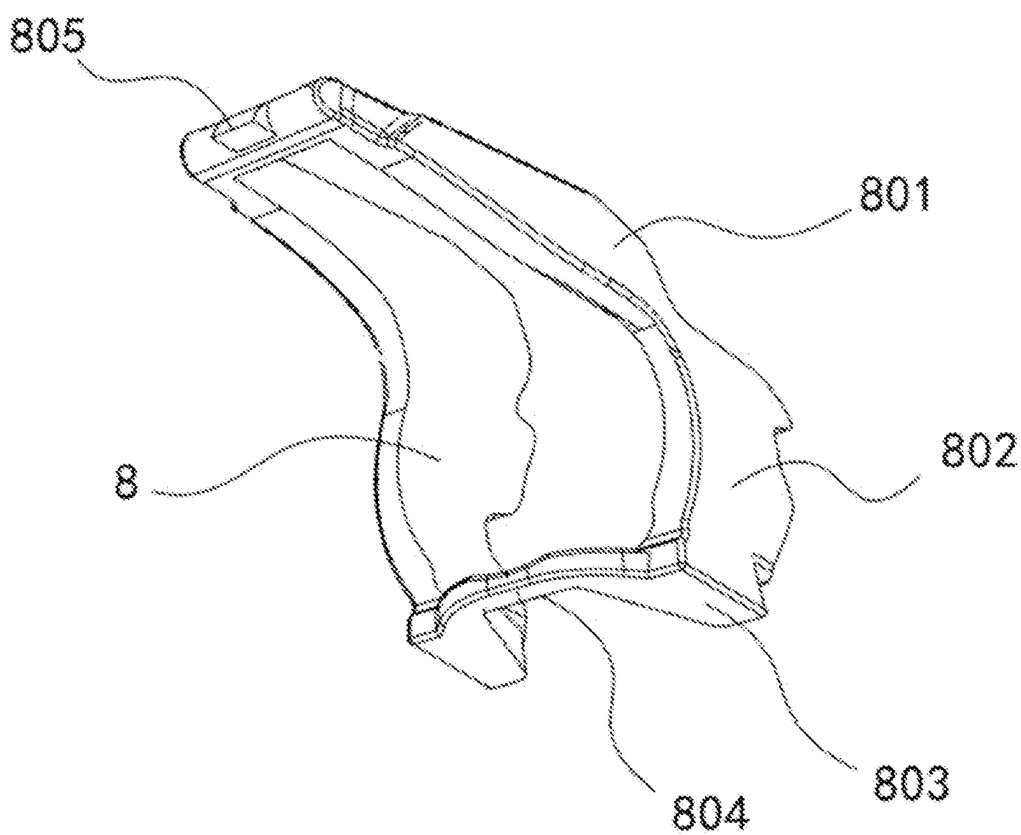


FIG. 14

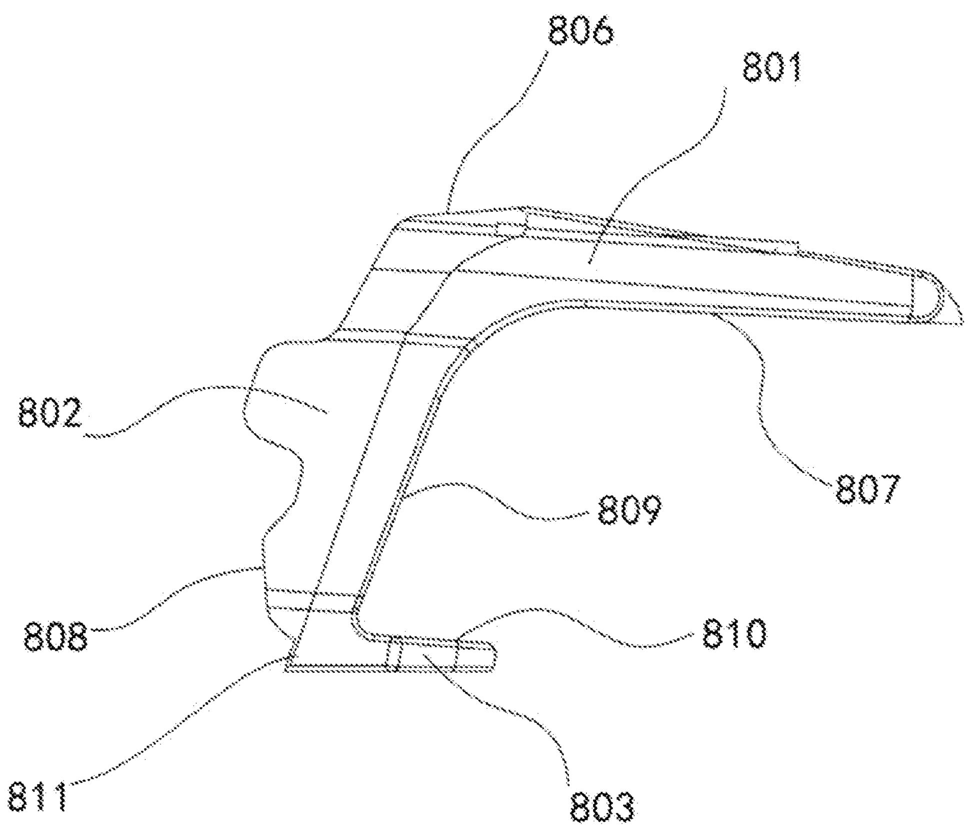


FIG. 15

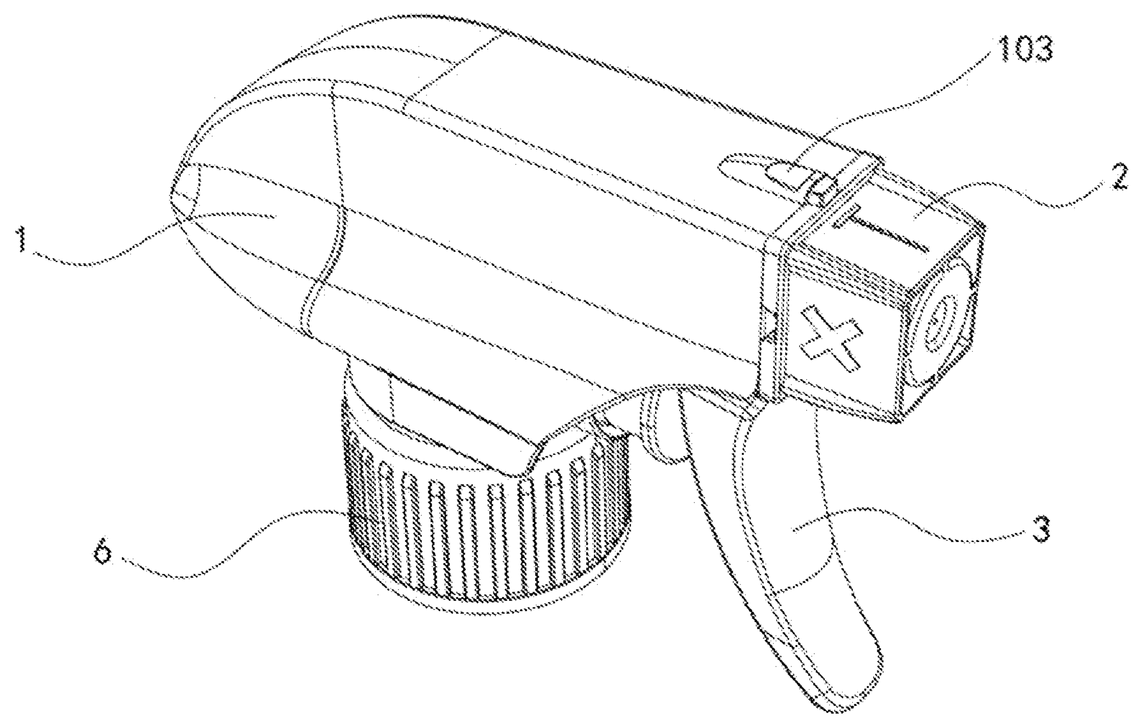


FIG. 16

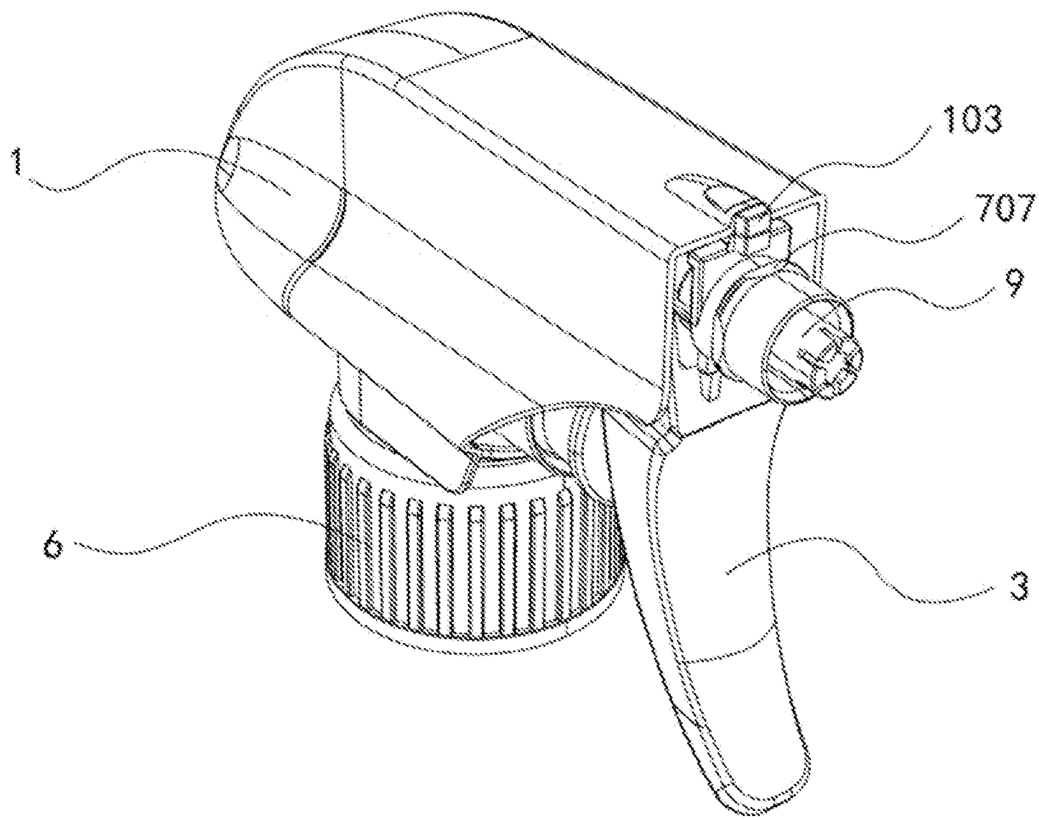


FIG. 17

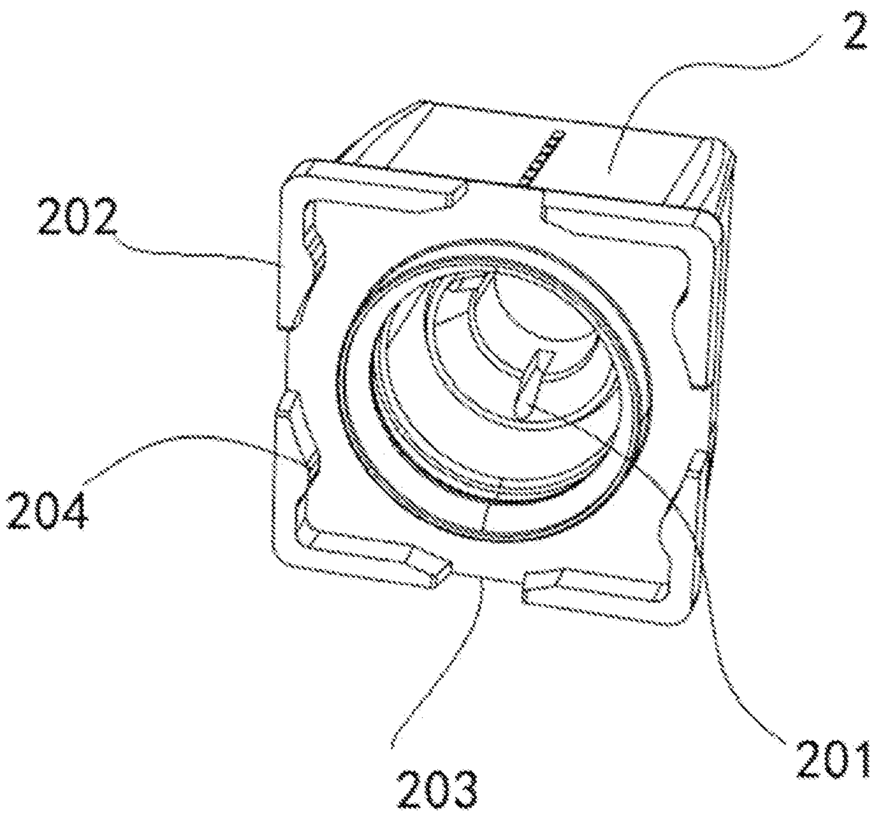


FIG. 18

ALL-PLASTIC SPRAY GUN

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of International Application No. PCT/CN2023/115077, with an international filing date of Aug. 25, 2023, which is based upon and claims priority to Chinese Patent Application No. 202311044093.3, filed on Aug. 17, 2023, the entire contents of all of which are incorporated herein by reference.

TECHNICAL FIELD

The present invention relates to the technical field of spray guns, and particularly relates to an all-plastic spray gun.

BACKGROUND

A spray gun mainly applied to cleaning and greening in life is a spraying device, which sprays water or a liquid in a bottle in the forms of mist, straight line and foam after a suction tube sucks the water.

Some existing sprayers for bottles substantially take a metal spring as power. A glass bead used as a water division device makes the liquid corrode the above two accessories, which indirectly causes slight injury to human body. The spray gun is neither environmental-friendly nor healthy to use, is difficult to recover, and cannot be recycled. The spray gun without a conventional metal spring and the glass bead is of an all-plastic structure, and the whole spray gun is only made from two materials, which can be integrally recycled. At present, although some sprayers are capable of achieving a continuous pressure function to achieve an injection purpose, a continuous pressure assembly for achieving injection is relatively complicated, and is formed by matching a piston, a spring, a longitudinal check valve, and the like. The above parts all are independent split parts, so that the assembling process of a product is relatively tedious.

BRIEF SUMMARY OF THE INVENTION

To solve the above technical problems, the present invention provide an all-plastic spray gun, including a housing, wherein the housing is connected to a gun body, a front end of the gun body is connected to a header cap, the header cap is connected to a water division head body, the water division head body is inserted into the gun body, the gun body is hinged to a trigger, the trigger is connected to a piston, the piston is slidably connected to a negative pressure column, an upper end of the negative pressure column is provided with a plastic spring, the plastic spring is sleeved outside the gun body, a bottom end of the gun body is connected to a blank cap, the blank cap is connected to a large ring, a perforative suction tube is inserted into the blank cap, the suction tube is connected to the gun body, the gun body is provided with a water outlet hole corresponding to the suction tube, and the gun body is internally provided with a glass bead in the water outlet hole.

Preferably, an upper end surface of a bottom of the gun body is provided with a spacing ring, the spacing ring is arranged to be semicircular-shaped, and the gun body is provided with a buckle located at a side edge of the spacing ring.

Preferably, an inner side of the trigger is provided with a first slot for mounting a piston rod, the piston rod is connected to the piston, and a position, located above the first slot, of the trigger is provided with a second slot for connecting the plastic spring.

Preferably, an inner wall at a top of the housing is provided with an insertion groove corresponding to the gun body, an inner wall at a bottom of the housing is provided with a fixture block, and the fixture block is provided with an arc-shaped groove.

Preferably, the gun body is provided with a clamp, the clamp includes a hoop and a support bar, the hoop is provided with an opening, and a side wall of the hoop is fixedly connected to the support bar.

Preferably, a top of the large ring is provided with an annular flange, and a gasket is mounted inside the large ring below the flange.

Preferably, the cover body can be matched with an open position at a rear end of the water division head body through the buckle, and the water division head body is in interference fit with the spring water division rod.

Preferably, the plastic spring includes a first support portion and a second support portion, a joint between upper ends of the first support portion and the second support portion is provided with an insertion block for being inserted into a second slot, bottom ends of the first support portion and the second support portion are connected to a bottom support portion, and the bottom support portion is provided with a groove.

Preferably, a lock piece is mounted on an upper end surface of a joint between the housing and the header cap.

Preferably, a position, close to a water outlet, of an inner wall of the header cap is provided with a liquid inlet groove, an edge of an end surface where the header cap contacts with a bearing rib is provided with four bosses, the four bosses are located at corners of the header cap, and a limit groove is formed between two adjacent bosses.

The present invention has the following technical effects and advantages:

1. In the present invention, by adopting the plastic spring to play a supporting role, the spray gun can be integrally recovered, a situation that a metal spring is used as power is avoided, and pollution of a liquid as the spring is corroded by the liquid is also avoided.

2. In the present invention, the trigger pushes the piston to compress water into the gun body, and in this case, the glass bead in the blank cap droops to seal the blank cap so as to prevent the water from flowing backwards. The liquid flows upwards to the spring water division rod to push the spring water division rod to compress, and the spring water division rod is elastically reset to push out the liquid to generate a continuous pressure, so that vaporific mist is generated at a high pressure and a high speed through the header cap.

3. In the present invention, the spring water division rod is mounted in the water division head body, and is then fixed by the cover body, so that the spray gun features simple structure and low mounting cost.

3. In the present invention, the lock piece is matched with the limit groove, so that the header cap has a locking function and is not easily rotated, and therefore, the safety is improved.

4. In the present invention, the arranged gasket and flange play a protecting role to the bottle opening to prevent a situation that the bottle opening is damaged as the spray gun collides with other objects in transportation.

5. In the present invention, the arranged clamp prevents a situation that the liquid leaks at the position where the piston is matched with the gun body as the trigger collides with other objects in transportation.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a structural schematic diagram of an all-plastic spray gun provided by an embodiment of the application.

FIG. **2** is an exploded view of the all-plastic spray gun provided by the embodiment of the application.

FIG. **3** is a sectional view of the all-plastic spray gun provided by the embodiment of the application.

FIG. **4** is a structural schematic diagram of a housing in the all-plastic spray gun provided by an embodiment of the application.

FIG. **5** is a structural schematic diagram of a clamp in the all-plastic spray gun provided by an embodiment of the application.

FIG. **6** is a structural schematic diagram of a large ring in the all-plastic spray gun provided by an embodiment of the application.

FIG. **7** is a sectional view of the large ring in the all-plastic spray gun provided by the embodiment of the application.

FIG. **8** is a structural schematic diagram of a water division head body in the all-plastic spray gun provided by an embodiment of the application.

FIG. **9** is a structural schematic diagram of a rear end of the water division head body in the all-plastic spray gun provided by an embodiment of the application.

FIG. **10** is a structural schematic diagram of a spring water division rod in the all-plastic spray gun provided by an embodiment of the application.

FIG. **11** is a structural schematic diagram of a rear end of the spring division rod in the all-plastic spray gun provided by an embodiment of the application.

FIG. **12** is a structural schematic diagram of a cover body in the all-plastic spray gun provided by an embodiment of the application.

FIG. **13** is a front view of the all-plastic spray gun provided by the embodiment of the application.

FIG. **14** is a structural schematic diagram of a plastic spring in the all-plastic spray gun provided by an embodiment of the application.

FIG. **15** is a side view of the plastic spring in the all-plastic spray gun provided by the embodiment of the application.

FIG. **16** is a structural schematic diagram of an all-plastic spray gun provided by another embodiment of the application.

FIG. **17** is a structural schematic diagram of a housing in the all-plastic spray gun provided by another embodiment of the application.

FIG. **18** is a structural schematic diagram of a header cap in the all-plastic spray gun provided by another embodiment of the application.

In the drawings, **1**, housing; **101**, insertion groove; **102**, fixture block; **103**, lock piece; **2**, header cap; **201**, liquid inlet groove; **202**, boss; **203**, limit groove; **204**, convex surface; **3**, trigger; **301**, first slot; **302**, second slot; **4**, piston; **401**, piston rod; **5**, clamp; **501**, hoop; **502**, support bar; **6**, large ring; **601**, flange; **602**, gasket; **7**, gun body; **701** water outlet hole; **702**, communication hole; **703**, buckle; **704**, arc-shaped hole; **705**, strip hole; **706**, spacing ring; **707**, bearing rib, **708**, clamping strip; **709**, water inlet hole; **8**, plastic spring; **801**, first support portion; **802**, second support portion; **803**, bottom support portion; **804**, groove; **805**, insertion block; **806**, first end surface; **807**, second end surface; **808**, third end surface; **809**, fourth end surface; **810**, fifth end surface; **811**, lug boss; **9**, water division head body; **91**, anterior segment portion; **92**, posterior segment portion; **93**, first opening; **94**, second opening; **95**, shoulder; **96**, convex rib; **97**, continuous pressure cavity; **10**, blank cap; **11**, suction tube; **12**, glass bead; **13**, cover body; **131**, positioning column, **14**, spring water division rod; **141**, positioning seat; **142**, sealing portion; **143**, elastic arm; **144**, seal ring; **145**, cambered surface; **146**, guide pillar; **147**, positioning hole.

DETAILED DESCRIPTION OF THE INVENTION

The present invention will be further described in detail below in combination with drawings and specific implementations. Embodiments of the present invention are provided for the purpose of illustration and description, and are not exhaustive or not limit the present invention to the disclosed form. Many modifications and variations will be apparent to those of ordinary skill in the art. The embodiment is chosen and described to better describe the principles and practical application of the present invention, and to enable those of ordinary skill in the art to understand the present invention and then design various embodiments with various modifications suitable to specific uses.

Referring to FIGS. **1**-**3**, in the embodiment, provided is an all-plastic spray gun, including a housing **1**, wherein the housing **1** is connected to a gun body **7**, a front end of the gun body **7** is connected to a header cap **2**, the header cap **2** is connected to a water division head body **9**, a spring water division rod **14** is inserted into the water division head body **9**, a cover body **13** is inserted into the spring water division rod **14**, the water division head body **9** is inserted into the gun body **7**, the gun body **7** is hinged to a trigger **3**, the trigger **3** is connected to a piston **4**, the piston **4** is slidably connected to a negative pressure column of the gun body **7**, the trigger **3** is pressed by a finger to drive the piston **4** to move along the negative pressure column of the gun body **7**, an upper end of the negative pressure column is provided with a plastic spring **8**, the plastic spring **8** is sleeved outside the gun body **7**, one end of the plastic spring **8** is connected to the trigger **3**, a bottom end of the gun body **7** is connected to a blank cap **10**, the blank cap **10** is connected to a large ring **6**, a perforative suction tube **11** is inserted into the blank cap **10**, the suction tube **11** is connected to the gun body **7**, the gun body **7** is provided with a water outlet hole **701** corresponding to the suction tube **11**, and the gun body **7** is internally provided with a glass bead **12** in the water outlet hole **701**. The glass bead **12** is located at an upper end of the suction tube **11**, a bottom of the water outlet hole **701** is communicated with a communication hole **702**, the communication hole **702** is communicated with the negative pressure column, the gun body **7** is provided with a water inlet hole **709** below the communication hole **702**, an upper end surface of a bottom of the gun body **7** is provided with a spacing ring **706**, the spacing ring **706** is arranged to be semicircular-shaped, the gun body **7** is provided with a buckle **703** located at a side edge of the spacing ring **706**, the buckle **703** is matched with the housing **1** to prevent the housing **1** from being separated from the buckle **703**, and the spring water division rod is mounted in the water division head body, and then is fixed by the cover body, so that the spray gun features simple structure and low mounting cost.

Further, an inner side of the trigger **3** is provided with a first slot **301** for mounting a piston rod **401**, the piston rod **401** is connected to the piston **4**, and a position, located above the first slot **301**, of the trigger **3** is provided with a second slot **302** for connecting the plastic spring **8**.

The trigger **3** is pressed by a finger to extrude the plastic spring **8** to deform and to drive the piston **4** to enter the negative pressure column. The trigger **3** is loosened, so that the plastic spring **8** is reset to push the trigger **3** to reset. A liquid enters the negative pressure column from the suction tube **11** and the water inlet hole **709**. The trigger **3** is pressed, so that the piston **4** enters the negative pressure column to extrude the liquid to enter the water outlet hole **701** from the communication hole **702**. In this case, the glass bead **12** in the blank cap **10** droops to seal the blank cap **10** to prevent the liquid from flowing backwards. The liquid flows upwards to the position of the water division head body **9** and is sprayed out in a fine waterway and a rotary waterway through the header cap **2**.

In the embodiment, referring to FIG. **2** and FIG. **4**, an inner wall at a top of the housing **1** is provided with an insertion groove **101** corresponding to the gun body **7**, an inner wall at a bottom of the housing **1** is provided with a fixture block **102**, the fixture block **102** is provided with an arc-shaped groove, and the fixture block **102** is inserted into a gap between the buckle **703** and the gun body **7** to prevent a displacement between the bottom of the housing **1** and the gun body **7**. The top of the gun body **7** is provided with a clamping strip **708**, the clamping strip **708** is inserted into the insertion groove **101**, and a tail end of the clamping strip **708** is provided with a boss. When the clamping strip **708** is inserted into the insertion groove **101**, the tail end of the clamping strip **708** is inserted into a tail end of the insertion groove **101**, and the boss fits the tail end of the insertion groove **101**. Meanwhile, an inner wall of the arc-shaped groove at the fixture block **102** clings to an outer wall of the gun body **7**, so that the housing **1** and the gun body **7** are mounted to prevent the gun body **7** from loosening.

In the embodiment, referring to FIG. **5**, the gun body **7** is provided with a clamp **5**, the clamp **5** includes a hoop **501** and a support bar **502**, the hoop **501** is provided with an opening, and a side wall of the hoop **501** is fixedly connected to the support bar **502**. A side section of the support bar **502** is in a reversed T shape. During mounting, one end of the support bar **502** fits the inner wall of the trigger **3** and the other end of the support bar **502** fits the outer wall of the gun body **7**. The hoop **501** clamps the outer side of the negative pressure column to prevent a situation that the liquid leaks at the position where the piston **4** is matched with the gun body **7** as the trigger **3** collides with other objects.

In the embodiment, referring to FIG. **6** and FIG. **7**, a top of the large ring **6** is provided with an annular flange **601**, the flange **601** is in a reversed conical shape, and a gasket **602** is mounted inside the large ring **6** below the flange **601**. The gasket **602** is circular. After the large ring **6** is connected to the gun body **7**, an upper end surface of the flange **601** clings to the bottom end of the gun body **7** to prevent liquid leakage at the large ring **6**. The arranged gasket **602** and flange **601** play a protecting role to the bottle opening to prevent a situation that the bottle opening is damaged as the spray gun collides with other objects in transportation.

In the embodiment, referring to FIGS. **8**-**12**, the cover body **13** can be matched with an open position at a rear end of the water division head body **9** through the buckle to effectively prevent the continuous pressure function from being affected as the liquid enters the spring water division rod **14**. The water division head body **9** is integrally columnar, and is internally provided with a continuous pressure cavity **97** where the spring water division rod **14** is inserted into. The water division head body **9** is provided with a shoulder **95** at a joint between an anterior segment portion **91** and a posterior segment portion **92**. The shoulder **95** divides the water division head body **9** into the anterior segment portion **91** located in the front portion and the posterior segment portion **92** located in the rear portion. An end of the anterior segment portion **91** is provided with a division channel for gathering the liquid and spraying the liquid out from the header cap **2**. Second openings **94** are formed at an interval in the anterior segment portion **91** of the water division head body **9** in a circumferential direction. Each of the second openings **94** forms a liquid outlet communicated with the water outlet hole **701**. First openings **93** are formed at an interval in the posterior segment portion **92** of the water division head body **9** in the circumferential direction. Each of the first openings **93** forms a liquid inlet communicated with the continuous pressure cavity **97**. An end of the posterior segment portion **92** is provided with a convex rib **96** for being mounted with the buckle of the cover body **13** in a fit manner.

The spring water division rod **14** includes a positioning seat **141**, wherein the positioning seat **141** is provided with a positioning hole **147**, the cover body **13** is provided with a positioning column **131** corresponding to the positioning hole **147**, and the positioning column **131** is inserted into the positioning hole **147**. After the cover body **13** is connected to the convex rib **96**, the spring water division rod **14** is mounted in the water division head body **9**. A side end of the positioning seat **141** is connected to an elastic arm **143** which is in an hourglass shape, and the other end of the elastic arm **143** is connected to a sealing portion **142**. An outer edge of the sealing portion **142** is provided with a seal ring **144**. The spring water division rod **14** is inserted into the continuous pressure cavity **97** to play a closing role through the seal ring **144**. A front end of the sealing portion **142** is provided with a cambered structure **145** in close fit with an inner conical surface of the water division head body **9**. A front end of the cambered structure **145** is further provided with a guide pillar **146** extending forwards. The spring water division rod **14** is in clearance fit with the water division head body **9** to effectively prevent the continuous pressure function from being affected as the liquid enters the water division head body **9**.

In the embodiment, referring to FIG. **13**, the gun body **7** is provided with an arc-shaped hole **704** and a strip hole **705** at the opening of the water outlet hole **701**, which facilitates positioning and mounting of the water division head body **9**.

In the embodiment, referring to FIG. **14** and FIG. **15**, the plastic spring **8** includes a first support portion **801** and a second support portion **802**, wherein a joint between upper ends of the first support portion **801** and the second support portion **802** is provided with an insertion block **805** for being inserted into a second slot **302**, bottom ends of the first support portion **801** and the second support portion **802** are connected to a bottom support portion **803**, the bottom support portion **803** is provided with a groove **804** clamped outside the gun body **7**, an upper end of the first support portion **801** is provided with a first end surface **806**, a lower end of the first support portion **801** is provided with a second end surface **807**, an outer side wall of the second support portion **802** is provided with a third end surface **808**, an inner side wall of the second support portion **802** is provided with a fourth end surface **809**, an upper end of the bottom support portion **803** is provided with a fifth end surface **810**, the plastic spring **8** is provided with a lug boss **811** on an inner side of the bottom support portion **803**, and the lug boss **811** is clamped at the spacing ring **706** to prevent a displacement generated at the bottom of the plastic spring **8**.

Specifically, an included angle between the first end surface **806** and the fifth end surface **810** is 5 degrees, with a deformation range being 45-35 degrees; an included angle between the second end surface **807** and the fourth end surface **809** is 109 degrees, with a deformation range being 60-140 degrees; an included angle between the third end surface **808** and the fifth end surface **810** is 95 degrees, with a deformation range being 45-125 degrees; and an included angle between the fourth end surface **809** and the fifth end surface **810** is 73 degrees, with a deformation range being 20-100 degrees.

By designing the included angles in the plastic spring **8**, the angle of deformation generated by pressing the plastic spring **8** by the trigger **3** is 8±20 degrees. When the trigger **3** is pressed, the plastic spring **8** is easy to deform, and when the trigger **3** is loosened, the plastic spring **8** is easy to reset.

In another embodiment, referring to FIGS. **16-18**, a lock piece **103** is mounted on an upper end surface of a joint between the housing **1** and the header cap **2**. An outer wall, close to the lock piece **103**, of the gun body **7** is provided with a bearing rib **707**. An inner wall of the header cap **2** is provided with an annular groove corresponding to the bearing rib **707**, so that the header cap **2** is capable of being rotatably connected to the gun body **7**. A position, close to a water outlet, of an inner wall of the header cap **2** is provided with a liquid inlet groove **201**. An edge of an end surface where the header cap **2** contacts with the bearing rib **707** is provided with four bosses **202**. The four bosses **202** are located at corners of the header cap **2**. A limit groove **203** is formed between two adjacent bosses **202**. A convex surface **204** is arranged on each of adjacent inner walls of the two bosses **202**. The convex surface **204** is matched with the bearing rib **707**.

In the embodiment, the lock piece **103** limits rotation of the header cap **2** to lock a nozzle. During unlocking, the upper end surface of the lock piece **103** is pressed, so that the lock piece **103** is separated from the header cap **2**. Then the header cap **2** is rotated to open the nozzle. The lock piece **103** is an elastic chuck and is matched with the limit groove **203**, so that the header cap **2** has the locking function and is not easily rotated, and therefore, the safety is improved.

The working principle of the present invention is as follows:

In the present invention, the header cap **2** has the locking function, so that the header cap **2** is not easily rotated, and therefore, the safety is improved. By adopting the plastic spring **8** to play a supporting role, the spray gun can be integrally recovered, the situation that the metal spring is used as power is avoided, and pollution of the liquid as the spring is corroded by the liquid is also avoided. The trigger pushes the piston to compress water into the gun body, and in this case, the glass bead in the blank cap droops to seal the blank cap so as to prevent the water from flowing backwards. The liquid flows upwards to the spring water division rod to push the spring water division rod to compress, and the spring water division rod is elastically reset to push out the liquid to generate a continuous pressure, so that vaporific mist is generated at a high pressure and a high speed through the header cap.

Apparently, the described embodiments are merely some rather than all of the embodiments of the present invention. Based on the embodiments of the present invention, all the other embodiments obtained by those of ordinary skill in the art and relevant arts without inventive effort are fall within the scope of the present invention. Structures, devices and operating methods not specifically described and explained in the present invention are implemented according to conventional means in the art unless otherwise specified and limited.

The invention claimed is:

1. An all-plastic spray gun, comprising a housing, wherein the housing is connected to a gun body, a front end of the gun body is connected to a header cap, the header cap is connected to a water division head body, a spring water division rod is inserted into the water division head body, a cover body is inserted into the spring water division rod, the water division head body is inserted into the gun body, the gun body is hinged to a trigger, the trigger is connected to a piston, the piston is slidably connected to a negative pressure column of the gun body, an upper end of the negative pressure column is provided with a plastic spring, the plastic spring is sleeved outside the gun body, a bottom end of the gun body is connected to a blank cap, the blank cap is connected to a large ring, a perforative suction tube is inserted into the blank cap, the suction tube is connected to the gun body, the gun body is provided with a water outlet hole corresponding to the suction tube, and the gun body is internally provided with a glass bead in the water outlet hole.

2. The all-plastic spray gun of claim **1**, wherein an upper end surface of a bottom of the gun body is provided with a spacing ring, the spacing ring is arranged to be semicircular-shaped, and the gun body is provided with a buckle located at a side edge of the spacing ring.

3. The all-plastic spray gun of claim **2**, wherein an inner side of the trigger is provided with a first slot for mounting a piston rod, the piston rod is connected to the piston, and a position, located above the first slot, of the trigger is provided with a second slot for connecting the plastic spring.

4. The all-plastic spray gun of claim **3**, wherein an inner wall at a top of the housing is provided with an insertion groove corresponding to the gun body, an inner wall at a bottom of the housing is provided with a fixture block, and the fixture block is provided with an arc-shaped groove.

5. The all-plastic spray gun of claim **1**, wherein the gun body is provided with a clamp, the clamp comprises a hoop and a support bar, the hoop is provided with an opening, and a side wall of the hoop is fixedly connected to the support bar.

6. The all-plastic spray gun of claim **1**, wherein a top of the large ring is provided with an annular flange, and a gasket is mounted inside the large ring below the flange.

7. The all-plastic spray gun of claim **1**, wherein the cover body can be matched with an open position at a rear end of the water division head body through the buckle, and the water division head body is in interference fit with the spring water division rod.

8. The all-plastic spray gun of claim **1**, wherein the plastic spring comprises a first support portion and a second support portion, a joint between upper ends of the first support portion and the second support portion is provided with an insertion block for being inserted into a second slot, bottom ends of the first support portion and the second support portion are connected to a bottom support portion, and the bottom support portion is provided with a groove.

9. The all-plastic spray gun of claim **1**, wherein a lock piece is mounted on an upper end surface of a joint between the housing and the header cap.

10. The all-plastic spray gun of claim **9**, wherein a position, close to a water outlet, of an inner wall of the header cap is provided with a liquid inlet groove, an edge of an end surface where the header cap contacts with a bearing rib is provided with four bosses, the four bosses are located at corners of the header cap, and a limit groove is formed between two adjacent bosses.

* * * * *